Invariant correlation: a general and straightforward method to quantify invariance in time-varying neural signals

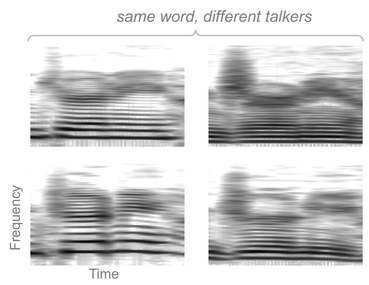
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1. Introduction

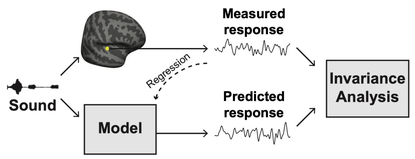
- Recognizing information in natural stimuli is challenging because different instances of the same feature differ substantially in their sensory characteristics
- Requires coding information in a way that is invariant or tolerant to identity-preserving changes in the signal



*We introduce the **invariant correlation**, a straightforward method to directly quantify invariance in time-varying neural signals from any sensory domain or recording modality*

2. Approach

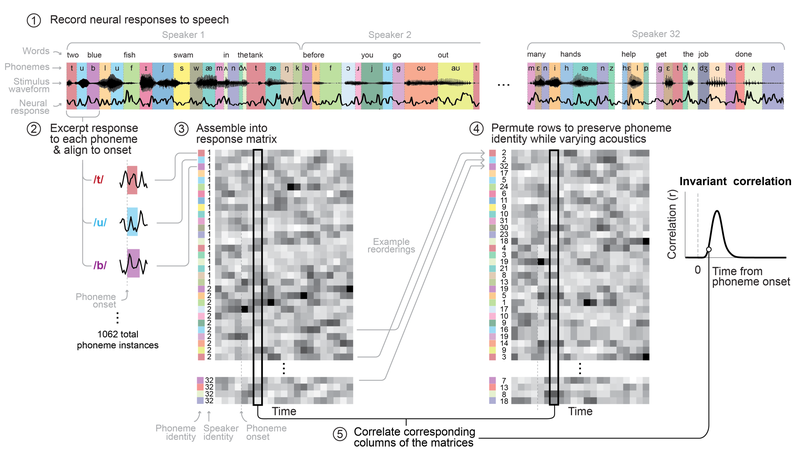
1. Use method to characterize organization of invariance across cortex
2. Repeat experiments on computational models and make direct brain-model comparisons



Illustrate approach using intracranial recordings from human auditory cortex to measure invariance to acoustic variation in neural representations of phonemes

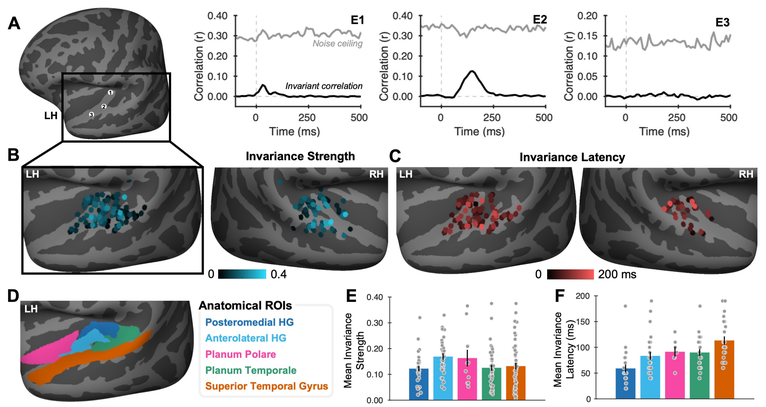
3. The invariant correlation

Measure invariance by quantifying the similarity of neural responses across different instances of stimuli sharing the same feature



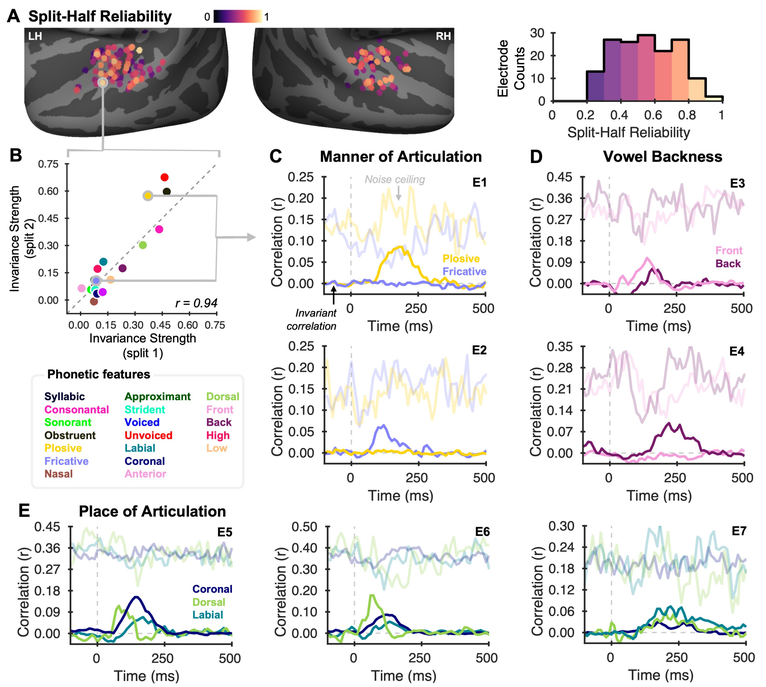
4. Measuring the strength and timing of invariance

Our method reveals substantial variability in invariance strength & latency



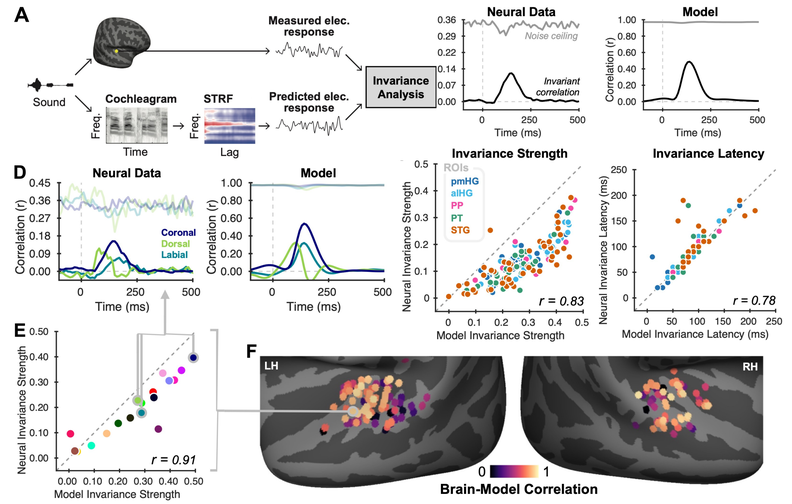
5. Identifying feature-specific patterns of invariance

Our method identifies structured patterns of invariance across phonetic features



6. Evaluating invariance in computational models

STRF encoding model reproduced many empirical patterns, suggesting much of the observed structure may reflect differences in underlying spectrotemporal tuning



7. Summary

- Introduce a general method to quantify the strength and temporal dynamics of invariance in neural responses and computational models
- Provides a continuous and interpretable measure of invariance, along with a noise ceiling, allowing values to be interpreted on a common scale and compared across recording sites
- Does not rely on assumptions about the stimulus-response mapping (e.g., encoding models) or readout scheme (e.g., decoding models)
- Flexible tool for probing invariance across sensory modalities, stimulus dimensions, and computational models

8. Acknowledgements

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